RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models

Cheng Niu[1], **Yuanhao Wu**[1], **Juno Zhu**[1], **Siliang Xu**[1], **Kashun Shum**[1], **Randy Zhong**[1], **Juntong Song**[1], and **Tong Zhang**[2]

[1]NewsBreak
[2]University of Illinois Urbana-Champaign
cheng.niu@newsbreak.com

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Abstract

Retrieval-augmented generation (RAG) has become a main technique for alleviating hallucinations in large language models (LLMs). Despite the integration of RAG, LLMs may still present unsupported or contradictory claims to the retrieved contents. In order to develop effective hallucination prevention strategies under RAG, it is important to create benchmark datasets that can measure the extent of hallucination. This paper presents RAGTruth, a corpus tailored for analyzing word-level hallucinations in various domains and tasks within the standard RAG frameworks for LLM applications. RAGTruth comprises nearly 18,000 naturally generated responses from diverse LLMs using RAG. These responses have undergone meticulous manual annotations at both the individual case and word levels, incorporating evaluations of hallucination intensity. We not only benchmark hallucination frequencies across different LLMs, but also critically assess the effectiveness of several existing hallucination detection methodologies. We show that using a high-quality dataset such as RAGTruth, it is possible to finetune a relatively small LLM and achieve a competitive hallucination detection performance when compared to the existing prompt-based approaches using state-of-the-art LLMs such as GPT-4. Furthermore, the fine-tuned model can effectively mitigate hallucination in LLM responses.

QUESTION: how to prepare to get an ultrasound?
REFERENCE: Passage 1: 1 If you are just interested in checking the gender, you can go as early as 17 weeks. 2 Any earlier, and the results may not be accurate, leading to an extra coat of paint for your bedroom. 3 To get good pictures, 3D ultrasounds are best performed between 20 and 32 weeks, and the best pictures are between 24 and 30 weeks.
...
RESPONSE: Based on the given passages, here's how to prepare for an ultrasound:
* For a gender reveal ultrasound, wait until at least 17 weeks of pregnancy to ensure accurate results.
* For 3D ultrasounds, schedule the appointment between 20 and 32 weeks of pregnancy for the best pictures ...
ANNOTATION
Span: between 20 and 32 weeks of pregnancy for the best pictures
Type: Evident Conflict
Reason: Original: "the best pictures are between 24 and 30 weeks", Generative: "between 20 and 32 weeks of pregnancy for the best pictures"

Table 1: An example of RAGTruth data from the question answering task. It contains context, response generated by LLM with and span-level annotation.

1 Introduction

Large language models (LLMs) have achieved remarkable success in a variety of tasks, including text generation (Li et al., 2022), machine translation (Kocmi and Federmann, 2023), and question answering (Zhao et al., 2023). However, one of the key challenges in deploying LLMs in real-world applications is their tendency to hallucinate (Kaddour et al., 2023). Hallucination in the context of LLMs usually refers to a situation where the model generates content that is not based on factual or accurate information (Rawte et al., 2023). The occasional generation of outputs that appear plausible but are factually incorrect significantly undermine the reliability of LLMs in real-world scenarios, such as medical diagnoses (Pal et al., 2023) and news summarization (Shen et al., 2023).

To reduce hallucination, various methods have been developed that can be applied at different stages of LLM lifecycle, including pre-training (Brown et al., 2020), supervised fine-tuning (Zhou et al., 2023; Zhang et al., 2023a), RLHF (Ouyang et al., 2022; Lin et al., 2022), and inference (Dhuliawala et al., 2023; Gao et al., 2023). In terms of detection, methods are developed by examining the model's intrinsic state (Guo et al., 2017), comparing it with external data and tools (Chern et al., 2023), or leveraging the

LLM's inherent powerful capabilities for self-checking (Agrawal et al., 2023; Manakul et al., 2023). Retrieval-augmented generation (RAG) is extensively used to supply LLMs with updated, relevant knowledge, significantly mitigating hallucination (Varshney et al., 2023). Nevertheless, even with RAG and other enhancements, LLMs still produce statements that are either unfounded or contradict the information provided in the retrieved references (Shuster et al., 2021).

Despite the growing awareness of the hallucination phenomenon, the understanding of hallucination in LLMs is still in its early stages. One key challenge is the lack of high-quality, large-scale datasets specifically designed for hallucination detection. This issue is particularly acute in RAG settings. Due to the relatively low hallucination ratio, a substantial increase in annotation resources is needed. Existing datasets for LLM hallucination detection are predominantly synthesized (Li et al., 2023). For instance, in Liu and Liu (2023); Longpre et al. (2021), prompts conflicting with conventional knowledge are purposely generated to trigger hallucinations. While these approaches are efficient at generating hallucinations, the resulting artificial hallucinations can substantially differ from those that naturally occur. In Chen et al. (2023); Hu et al. (2023), hallucination datasets are developed by manual annotations of naturally produced LLM responses. However, these datasets are of limited size and are not specifically focused on the RAG scenario.

In this paper, we introduce a large-scale high-quality dataset specifically designed for word-level hallucination detection for RAG applications. Using this dataset, we have conducted an extensive benchmarking of mainstream LLMs to assess their tendency to generate hallucinations, as well as evaluate current methods for hallucination detection. Additionally, we have demonstrated superior performance in identifying hallucinations by fine-tuning LLM with RAGTruth dataset. Our key contributions are:

(i) We propose RAGTruth, a large-scale word-level hallucination evaluation dataset specifically for the RAG scenario across several common tasks. It consists of nearly 18,000 fully annotated natural responses generated from major open-source and closed-source LLMs.

(ii) We perform a comprehensive comparison of different hallucination detection methods at both the passage and word levels.

(iii) We present a baseline method of fine-tuning LLM for hallucination detection. It is shown that by fine-tuning the Llama-2-13B model on the RAGTruth training data, we can achieve results competitive to the existing prompt-based approaches using GPT-4. This shows the potential of developing better hallucination detection methods using RAGTruth.

(iv) We show that by using our finetuned hallucination detector, it is possible to significantly reduce the occurrence of hallucinations in the responses from LLMs. The improvement holds even for models with inherently low hallucination rates, such as GPT-4.

2 Related Work

2.1 Hallucination of Large Language Models

Though hallucination in traditional natural language generation (NLG) contexts has been widely studied(Ji et al., 2023), comprehending and tackling this problem in the context of LLMs presents distinct challenges(Zhang et al., 2023b). Existing research has demonstrated that incorporating up-to-date, relevant knowledge in the prompt can effectively reduce fact-conflicting hallucination (Vu et al., 2023; Lewis et al., 2021). This approach, referred to as *Retrieval-Augmented Generation* (RAG), is widely used in real-world LLM applications. For instance, Google Bard [1] and Microsoft BingChat [2] have implemented this technique.

2.2 Hallucination Evaluation Datasets

Extensive research has focused on hallucination benchmarks within conventional Natural Language Generation settings (Dziri et al., 2022; Zhong et al., 2021; Durmus et al., 2020; Lin et al., 2022). With the rise of LLMs, the detection of hallucinations has become increasingly challenging, necessitating the development of high-quality datasets for LLM evaluation (Chen and Shu, 2023). Contributions in this domain include HaluEval (Li et al., 2023), which introduced datasets encompassing both synthetically and naturally generated LLM responses, and FELM (Chen et al., 2023), which concentrated on naturally generated LLM responses across multiple domain tasks. RefChecker (Hu et al., 2023), a

[1] https://bard.google.com
[2] https://www.bing.com

distinctive approach, breaks down claims in LLM responses into triples and utilizes human annotation to assess the truthfulness of facts. Notably, these works primarily focus on annotating factual hallucinations in LLM responses. Distinguishing from previous research, our work centers on the evaluation of LLMs within RAG settings.

2.3 Hallucination Detection Methods

Researchers have been exploring various methods to enhance the reliability of LLMs by detecting hallucinations. In Azaria and Mitchell (2023); Xiao and Wang (2021); Malinin and Gales (2021), intrinsic model uncertainty metrics such as token-level probability and entropy are used to detect hallucinations. When direct access to output uncertainty is not feasible, as in the case with limited APIs like GPT-4, an alternative approach involves employing a fully accessible LLM as a proxy (Manakul et al., 2023). In Falke et al. (2019); Barrantes et al. (2020), natural language inference modules are adapted to check the information consistency between the articles and their summaries, and it has been shown that external knowledge is helpful for detecting factual hallucinations. (Guo et al., 2022; Mallen et al., 2022). Additionally, methods that leverage the inherent capabilities of LLMs have been proposed for self-checking, such as verbalization-based and consistency-based methods (Xiong et al., 2023; Manakul et al., 2023). These techniques aim to detect hallucinations without relying on internal states or external data and tools.

3 Construction Process of RAGTruth

We established a data generation and annotation pipeline as shown in Figure 1.

3.1 Hallucination Taxonomy

Different from open-end generation, under RAG setting, the prompt contains rich context information, and the model is generally required to generate text based on the provided context. The detection and mitigation of inconsistencies between retrieved information and responses emerge as significant sources of hallucination.

As outlined below, we categorize the hallucination in the RAG setting into four types. For concrete examples of each type, please refer to Appendix A.

Evident Conflict: for when generative content presents direct contraction or opposition to the provided information. These conflicts are easily verifiable without extensive context, often involving clear factual errors, misspelled names, incorrect numbers, etc.

Subtle Conflict: for when generative content presents a departure or divergence from the provided information, altering the intended contextual meaning. These conflicts often involve substitution of terms that carry different implications or severity, requiring a deeper understanding of their contextual applications.

Evident Introduction of Baseless Information: for when generated content includes information not substantiated in the provided information. It involves the creation of hypothetical, fabricated, or hallucinatory details lacking evidence or support.

Subtle Introduction of Baseless Information: is when generated content extends beyond the provided information by incorporating inferred details, insights, or sentiments. This additional information lacks verifiability and might include subjective assumptions or commonly observed norms rather than explicit facts.

3.2 Response Generation

Tasks and Data Sources We selected three widely recognized generation tasks with RAG settings for response generation: Question Answering, Data-to-text Writing, and News Summarization.

For the task of question answering, we conducted a random sampling from the training set of MS MARCO (Nguyen et al., 2016). To reduce the difficulty of annotation, we selected only those questions related to daily life, and preserved only three retrieved passages for each question. Then we prompted LLMs to generate answers for each question solely based on the retrieved passages.

For the data-to-text writing task, we prompted LLMs to generate an objective overview for a randomly sampled business in the restaurant and nightlife categories from the Yelp Open Dataset (Yelp, 2021). In this dataset, information pertaining to a business is represented using structured data. To streamline the annotation process, we focused only on the following business information fields: *BusinessParking*, *RestaurantsReservations*, *OutdoorSeating*, *WiFi*, *RestaurantsTakeOut*, *RestaurantsGoodForGroups*, *Music*, and *Ambience*. In addition to the structured data, we have also included up to three business-related user reviews

Figure 1: Data gathering pipeline. Taking a data-to-text writing task as an example, our data gathering pipeline includes 2 steps: 1) response generation. We generated responses with multiple LLMs and natural prompts. 2) human annotation. Human labeler annotated hallucinated spans in LLM responses.

to enrich the context information. In the prompt, these information is represented in JSON format.

For the news summarization task, we randomly selected documents from the training set of the well-known CNN/Daily Mail dataset (See et al., 2017) as well as recent news articles from a prestigious news platform. LLMs were prompted to generate a summary for each of the source news.

Models The following six models with strong instruction-following ability are used for response generation: GPT-3.5-turbo-0613 and GPT-4-0613 from OpenAI (OpenAI, 2023); Mistral-7b-Instruct from Mistral AI (Jiang et al., 2023); Llama-2-7B-chat, Llama-2-13B-chat and Llama-2-70B-chat (4bit quantized)[3] from Meta (Touvron et al., 2023). To ensure a fair comparison, the prompts used for response generation are kept straightforward with subtle differences among various models to optimize their performance. We provide detailed prompts in the Appendix B.

For each sample, we collected one response from each model. As a result, we got a total of 6 responses for each input sample.

3.3 Human Annotation

Identifying AI-generated hallucinations is a challenging task. It requires a strong capacity for critical thinking to understand the logical flow of various texts, along with meticulous attention to detail for spotting subtle inaccuracies and inconsistencies. Moreover, a certain level of media literacy and knowledge of current affairs is crucial to grasp the subjects discussed in news-related sample data. Therefore, we chose annotators who are proficient in English and possess a bachelor's degree in English, Communications, or relevant fields to ensure the accuracy and reliability of the annotation results. We recruited annotators from a professional vendor and paid them at a rate of $25 per hour per individual.

The annotators are invited to perform annotation tasks using Label Studio (Tkachenko et al., 2020-2022). Each labeling task is presented within one page, comprising the following components: 1) the context provided to the AI models; 2) a set of 6 responses, generated by different AI models. Our annotation interface is available in Appendix C.

Their task was to annotate the specific spans of the generated text that contains hallucinated information and categorize them into the four types. To ensure the quality of the annotations, each response is independently labeled by two annotators. The consistency rate of two annotators was 91.8% at the response level and 78.8% at the span level. In cases where there is a considerable difference between the two annotations, a third review is undertaken.

3.4 Annotations for Adaptive Evaluation

In different contexts, the definition and criteria for hallucination vary, and the annotation of hallucination is not always straightforward. In contentious cases, additional annotations are provided to accurately reflect these situations. This approach enables users to adopt various evaluation strategies tailored to their specific application circumstances. Please refer to Appendix C for more statistical information about these annotations.

Implicit Truth The extensive world knowledge and ability of LLMs is a significant advantage in open-ended generation scenarios. But in the context of this paper, which focuses on the relatively strict RAG scenarios, we have labeled information that is not mentioned in the reference but may be truthful as hallucinations. For instance, mentioning

[3] https://huggingface.co/TheBloke/Llama-2-70B-Chat-AWQ

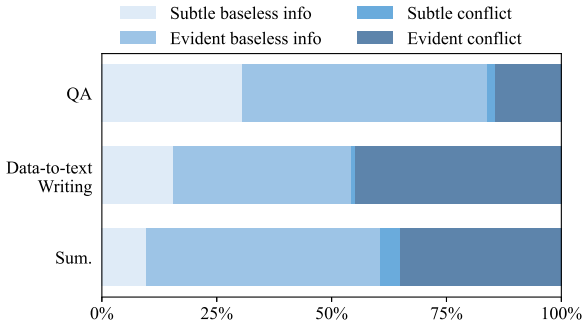

Figure 2: Frequency of different types of hallucination by task.

a local officer's name not present in the reference or claiming that a restaurant accepts credit card payments without any basis.

The decision is based on the observation that LLMs have a relatively high chance of making errors when generating detailed facts, partly because their embedded knowledge can be outdated. Therefore, RAG applications usually instruct LLMs not to generate factual content without the support of references. Besides, we provided an additional span-level annotation named *implicit_true* for these spans to accommodate different application needs.

Differences in Handling Null Value In the data-to-text writing task, certain fields sometimes are with null values. We observed that in the generated results, null is often interpreted as false by some models. Since the more common expressions for negation in our dataset are the boolean value *False* or the text *No*, we labeled these instances as hallucinations (evident introduction of baseless info) and provided a special span-level annotation named *due_to_null* for these spans. In the subsequent hallucination detection experiments, our prompts will be aligned with this standard.

4 Hallucination Benchmark Analysis

4.1 Basic Statistics

We presented detailed statistics of RAGTruth in Table 2. Compared to existing datasets for hallucination detection (Cao et al., 2023; Kamoi et al., 2023), the RAGTruth dataset is considerably large in scale. The corpus contains a total of 2,965 instances of data, which include 989 instances for question answering, 1,033 instances for date-to-text writing, and 943 instances for news summarization. Each instance comprises responses from 6 different models. As shown in Table 2, the RAGTruth dataset also features longer prompt and response lengths than existing datasets for hallucination detection (Wang et al., 2020).

4.2 Hallucination Statistics

Hallucination Types As shown in Figure 2, the generation of information baseless in the context was significantly more prevalent than the generation of information conflicting with the context, especially for the question answering tasks. Within the two major categories of *baseless info* and *conflict*, the more severe hallucinations, namely *Evident baseless info* and *Evident conflict*, respectively, account for a significant portion. This observation highlights the importance and challenges of LLMs hallucination mitigation, even in RAG settings.

Hallucination vs Tasks As shown in Table 2, across the three tasks, the date-to-text writing task exhibited the highest frequency of hallucinations in its responses. Inconsistent handling of JSON format data, especially time and attributes, contributed to a significant number of hallucinations in this task. Interestingly, the models did not show a higher rate of hallucinations for recent news compared to outdated news. This could be attributed to the shorter context length in the recent news subtask compared to the CNN/DM subtask.

Hallucination vs Models Table 3 illustrates that among the data we collected, OpenAI's two models demonstrated notably lower hallucination rates compared to others. Specifically, GPT-4-0613 exhibited the lowest hallucination frequency.

To more clearly compare the hallucination rate of different models, we calculated the hallucination density for each model across three tasks. Hallucination density is defined as the average number of hallucination spans per hundred words in the responses. In the Llama2 series, a clear negative correlation was observed between the model scale and hallucination density, aside from the data-to-text writing tasks. Despite its strong performance in various benchmarks and leaderboards (Zheng et al., 2023), the Mistral-7B-Instruct model generated the highest number of responses containing hallucinations.

Hallucination vs Length After removing the top and bottom 5% of outliers, we partitioned the data for each task type into three equal-sized groups according to the length of the context/response. We then computed the average number of hallucinated spans per response within each group. As shown

Task	# Instance	# Resp.	Context Length		Resp. Length		Hallucination		
			Mean	Max	Mean	Max	# Resp.	% Resp.	# Span
Question Answering	989	5934	243	509	119	381	1724	29.1%	2927
Data-to-text Writing	1033	6198	354	1253	159	369	4254	68.6%	9290
Summarization(CNN/DM)	628	3768	648	1749	124	632	1165	30.9%	1474
Summarization(Recent News)	315	1890	369	481	89	240	521	27.6%	598
Overall	2965	17790	381	1749	131	632	7664	43.1%	14289

Table 2: The basic statistics of RAGTruth. Here "Resp." stands for "Response".

Model	Question Answering			Data-to-text Writing			Summarization			Overall	
	# Resp.	# Span	Density	# Resp.	# Span	Density	# Resp.	# Span	Density	# Resp.	# Span
GPT-3.5-turbo-0613	75	89	0.12	272	384	0.18	54	60	0.05	401	533
GPT-4-0613	48	51	0.06	290	354	0.27	74	80	0.08	406	485
Llama-2-7B-chat	510	1010	0.59	888	1775	1.27	434	517	0.58	1832	3302
Llama-2-13B-chat	399	654	0.48	983	2803	1.53	295	342	0.41	1677	3799
Llama-2-70B-chat†	320	529	0.40	863	1834	1.15	212	245	0.26	1395	2608
Mistral-7B-Instruct	378	594	0.59	958	2140	1.51	617	828	0.86	1953	3562

Table 3: Hallucination counts and density of models. †: We used 4-bit quantized version of Llama-2-70B-chat.

CLB	Summarization	D2T Writing	QA
1	$0.29_{(176,368]}$	$1.51_{(178,273]}$	$0.50_{(131,187]}$
2	$0.36_{(368,587]}$	$1.48_{(273,378]}$	$0.51_{(187,288]}$
3	$0.44_{(587,1422]}$	$1.49_{(378,731]}$	$0.49_{(288,400]}$
RLB	**Summarization**	**D2T Writing**	**QA**
1	$0.34_{(44,87]}$	$1.20_{(93,131]}$	$0.21_{(19,93]}$
2	$0.32_{(87,119]}$	$1.59_{(131,175]}$	$0.37_{(93,138]}$
3	$0.44_{(119,245]}$	$1.69_{(175,258]}$	$0.87_{(138,257]}$

Table 4: Average number of hallucinations per response in different context length buckets (CLB) and response length buckets (RLB) for the three types of tasks. The subscript denotes the minimum and maximum length of this bucket.

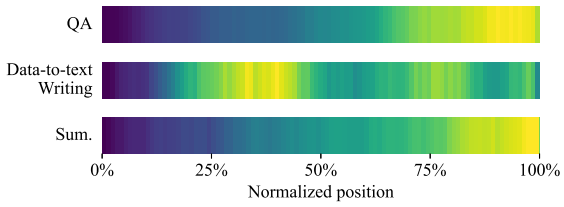

Figure 3: Heatmaps of normalized hallucination occurrence positions. The probability of hallucinations occurring is higher in brighter areas.

in Table 4, there is a clear overall trend of an increase in the average number of hallucinations as the response length grows. Only the average number of hallucinations in news summarization tasks significantly increases with the length of the context. This may be because the contexts in the other two tasks are more structured, and an increase in length does not significantly raise the difficulty of understanding the content.

Location of Hallucinations In Figure 3, we present the heatmap of the hallucination occurrence positions. Hallucinations are significantly more likely to occur towards the end of responses in question-answering and news summarization tasks. Compared to other tasks, the data-to-text writing task has a relatively higher occurrence of hallucinations in the first half. In that bright area, hallucinations concerning business attributes frequently occur.

5 Experimental Setup

5.1 Hallucination Detection Algorithms

Using RAGTruth, we conducted experiments with the following four distinct algorithms for hallucination detection:

Hallucination Detection Prompt: Hallucination detection prompts are manually crafted to instruct LLMs (GPT-4-turbo and GPT-3.5-turbo) in assessing whether a given reference-response pair contains hallucinated content and to identify the corresponding hallucinated spans in the response. For detailed information about these prompts, please refer to Appendix D.

SelfCheckGPT (Manakul et al., 2023): SelfCheckGPT employs a zero-resource, sampling-based method to fact-check the responses of black-box models. When processing each response in RAGTruth, 3 extra responses from the same model were sampled and served as references, and GPT-3.5-turbo was used to verify consistency. We detected hallucinations sentence-by-sentence within a response, and then aggregated these results to